INCH-POUND
MIL-DTL-9395/33D
24 September 2018
SUPERSEDING
MIL-DTL-9395/33C
3 October 2006

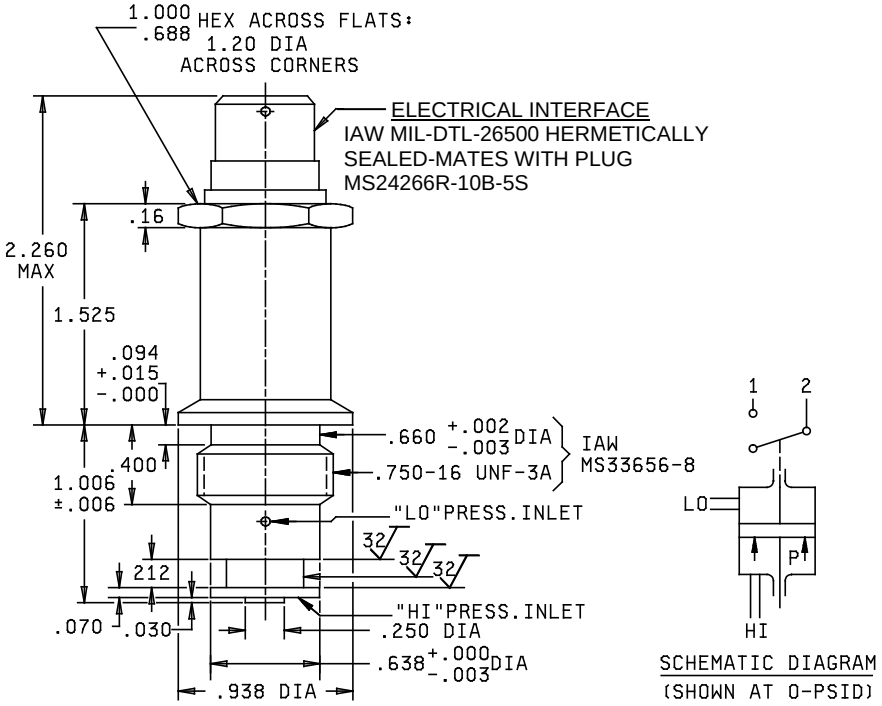
DETAIL SPECIFICATION SHEET

SWITCHES, PRESSURE, (DIFFERENTIAL), TYPE III, 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this document and MIL-DTL-9395.

Inactive for new design after 25 April 2013



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .005 (.13 mm) for three place decimals and .03 (.76 mm) for two place decimals.
4. MS24266R-10B-5S details are found MS24266. MS33656-8 details are found in SAE-AS4395.

FIGURE 1. Switch.

AMSC N/A

FSC 5930

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REQUIREMENTS:

Dimensions, configuration, and electrical schematic: See figure 1.

Weight: 0.14 pound, maximum.

Calibration: See tables I, II, III, IV, and V.

Proof pressure: 3,000 lb_f/in^2 high over low and simultaneously.

System pressure: 1,500 lb_f/in^2 high over low and simultaneously.

Burst pressure: 7,500 lb_f/in^2 high over low and simultaneously.

Electrical ratings:
Operating voltage: 28 V dc.
Current rating: 5 amperes resistive
3 amperes inductive.

Seal:
High pressure chamber: Media proof. Subject switches to proof pressure for 2 minutes using hydraulic fluid IAW MIL-PRF-6083 with chamber pressure continuously being monitored. Isolate the chamber at proof pressure with the chamber disconnected from the pressure source. Under that condition, the pressure shall not drop more than 1 lb_f/in^2 for the first 30 seconds to allow stabilization of test equipment; no pressure loss is allowed thereafter.

Low pressure chamber: Media proof. Tested as above except low pressure port is connected to high pressure port.

Electrical chamber: Unsealed.

Electrical connector: See figure 1.

Pressure ports: See figure 1.

Media: Hydraulic fluid IAW MIL- PRF-6083 or MIL-PRF-83282.

High temperature (operating and nonoperating): B (275°F).

Low temperature (operating and nonoperating): D (-65°F).

Altitude: C (70,000 feet).

Shock: C (100 G).

Vibration: S (test condition D of MIL-STD-202-204).

Life mechanical: A (100,000 cycles).

Life electrical: C (50,000 cycles).

Acceleration: C (8 G).

Pulsation amplitude: A (0%).

Pulsation frequency: A (0 Hz).

Pressure surge: B (more than 1,000 $lb_f/in^2/sec$).

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PART NUMBER: The part number consists of the prefix M9395/33- followed by five code characters, the first from table I (denoting setting mode), the second from table II, the third from table III (denoting high pressure setting), the fourth from table II, and the fifth from table III (denoting the low pressure setting). The following example illustrates the method used in generating a coded part number:

EXAMPLE: M9395/33- A 2 Y Q G

Prefix

Denotes the high pressure setting is "AT or "BY" and low pressure setting is "AT" or "BY"

Denotes the coarse high pressure setting is 250 lb_f/in^2

Denotes the fine high pressure setting to be added if 0 lb_f/in^2, and the high pressure setting is max (BY)

Denotes the coarse low pressure setting is 140 lb_f/in^2

Denotes the fine low pressure setting to be added is 0, and the tolerance is $\pm$15 lb_f/in^2.

M9395/33-A2YQG is the part number for a switch that is actuated on increasing pressure by 250 lb_f/in^2 and that is deactuated on decreasing pressure at 140 $\pm$15 lb_f/in^2.

Supersession data: M9395/33-01 is superseded by M9395/33-A2YQG.

TABLE I. Code for pressure setting modes.

Code character	Setting modes	
	High pressure	Low pressure
A	AT (or BY)	AT (or BY)
B	AT (or BY)	Differential
C	Differential	AT (or BY)

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TABLE II. Codes for coarse pressure setting (within lb_f/in^2).

Code	Coarse pressure setting (lb_f/in^2)	Code	Coarse pressure setting (lb_f/in^2)	Code	Coarse pressure setting (lb_f/in^2)
A	0	N	120	1	240
B	10	P	130	2	250
C	20	Q	140	3	260
D	30	R	150	4	270
E	40	S	160	5	280
F	50	T	170	6	290
G	60	U	180	7	300
H	70	V	190	8	310
J	80	W	200	9	320
K	90	X	210	0	340
L	100	Y	220		
M	110	Z	230		

TABLE III. Code for combination of fine pressure setting and tolerance. (This quantity should be added to coarse setting selected from table II).

Fine pressure setting (lb_f/in^2)	Code characters (tolerance lb_f/in^2)											
	Min	±5	±10	±15	±20	±25	±30	±35	±40	±50	±60	Max
0	A	C	E	G	J	L	N	Q	S	U	W	Y
5	B	D	F	H	K	M	P	R	T	V	X	Z

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced Documents:

MIL-PRF-6083
MIL-PRF-83282
MIL-STD-202-204
SAE-AS4395
MIL-DTL-9395
MIL-DTL-26500
MS24266

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2018-044)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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